

Classification Error Metrics



Objectives

- Understand the evaluation metrics
 - Accuracy
 - Precision
 - Recall / Sensitivity (True Positive Rate)
 - F1 score
 - Specificity (True Negative Rate)
 - ROC curve
 - AUC curve

Choosing the Right Error Measurement

- You are asked to build a classifier for leukemia
- Training data: 1% patients with leukemia, 99% healthy
- Measure accuracy: total % of predictions that are correct
- Build a simple model that always predicts "healthy"
- Accuracy will be 99%...

Accuracy

- metric for evaluating classification models
- fraction of predictions for our model to be right

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

For binary classification, accuracy can also be calculated in terms of positives and negatives as follows

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Accuracy

- Let's try calculating accuracy for the following model that classified 100 tumors as malignant (the positive class) or benign (the negative class):

True Positive (TP): • Reality: Malignant • ML model predicted: Malignant • Number of TP results: 1	False Positive (FP): • Reality: Benign • ML model predicted: Malignant • Number of FP results: 1
False Negative (FN): • Reality: Malignant • ML model predicted: Benign • Number of FN results: 8	True Negative (TN): • Reality: Benign • ML model predicted: Benign • Number of TN results: 90

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{1 + 90}{1 + 90 + 1 + 8} = 0.91$$

Accuracy comes out to 0.91, or 91% (91 correct predictions out of 100 total examples). That means our tumour classifier is doing a great job of identifying malignancies, right?

Analysis

Of the 100 tumours examples, 91 are benign (90 TNs and 1 FP) and 9 are malignant (1 TP and 8 FNs).

Of the 91 benign tumours, the model correctly identifies 90 as benign. That's good. However, of the 9 malignant tumours, the model only correctly identifies 1 as malignant—a terrible outcome, as 8 out of 9 malignancies go undiagnosed!

While 91% accuracy may seem good at first glance, another tumour-classifier model that always predicts benign would achieve the exact same accuracy (91/100 correct predictions) on our examples. In other words, our model is no better than one that has zero predictive ability to distinguish malignant tumours from benign tumours.

Accuracy alone doesn't tell the full story when you're working with a **class-imbalanced data set**, like this one, where there is a significant disparity between the number of positive and negative labels.

Precision

- Precision attempts to answer the following question:
 - What proportion of positive identifications was actually correct?
 - Identifying Only Positive Instances

$$Precision = \frac{TP}{TP + FP}$$

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Precision

- Let's try calculating accuracy for the following model that classified 100 tumors as malignant (the positive class) or benign (the negative class):

True Positive (TP): • Reality: Malignant • ML model predicted: Malignant • Number of TP results: 1	False Positive (FP): • Reality: Benign • ML model predicted: Malignant • Number of FP results: 1
False Negative (FN): • Reality: Malignant • ML model predicted: Benign • Number of FN results: 8	True Negative (TN): • Reality: Benign • ML model predicted: Benign • Number of TN results: 90

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{1}{1 + 1} = 0.5$$

Analysis

The model only predicts a tumour is malignant, it is correct 50% of the time

Recall

- Also called sensitivity, or true positive rate (TPR)
- Recall attempts to answer the following question:
 - What proportion of actual positives was identified correctly?
 - Identifying All Positive Instances

$$\text{Recall} = \frac{TP}{TP + FN}$$

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Recall

Let's try calculating accuracy for the following model that classified 100 tumors as malignant (the positive class) or benign (the negative class):

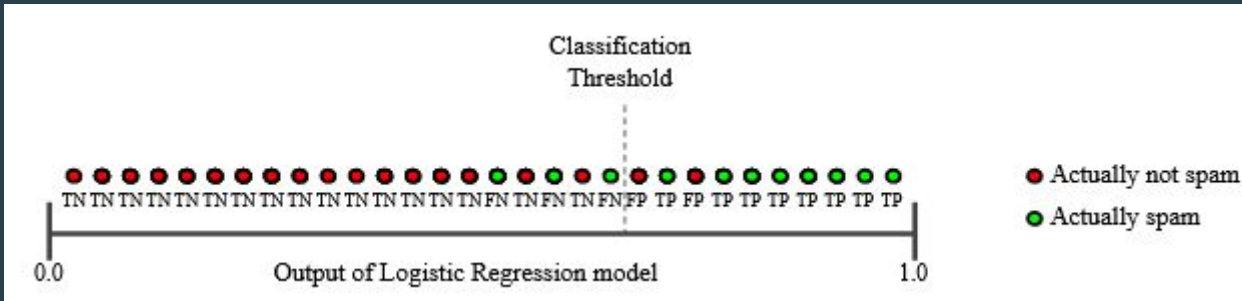
True Positive (TP): - Reality: Malignant - ML model predicted: Malignant - Number of TP results: 1	False Positive (FP): - Reality: Benign - ML model predicted: Malignant - Number of FP results: 1
False Negative (FN): - Reality: Malignant - ML model predicted: Benign - Number of FN results: 8	True Negative (TN): - Reality: Benign - ML model predicted: Benign - Number of TN results: 90

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{1}{1 + 8} = 0.11$$

Analysis

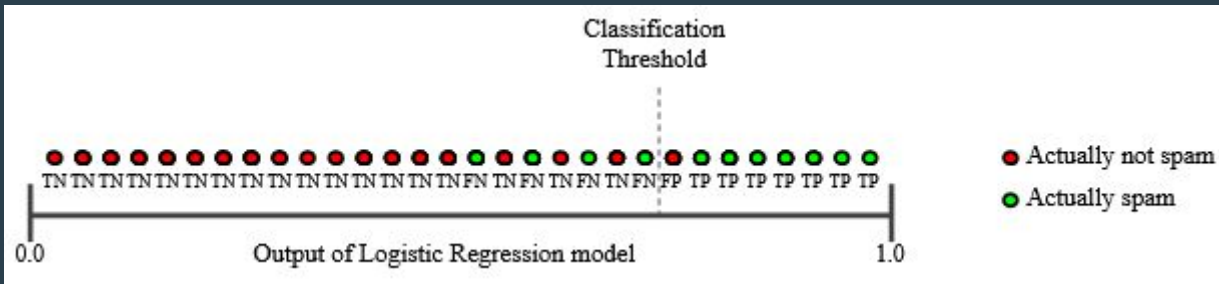
The model only correctly identifies 11% of all malignant tumours

Precision and Recall: A Tug of War



True Positives (TP): 8	False Positives (FP): 2
False Negatives (FN): 3	True Negatives (TN): 17

- effect of increasing the classification threshold



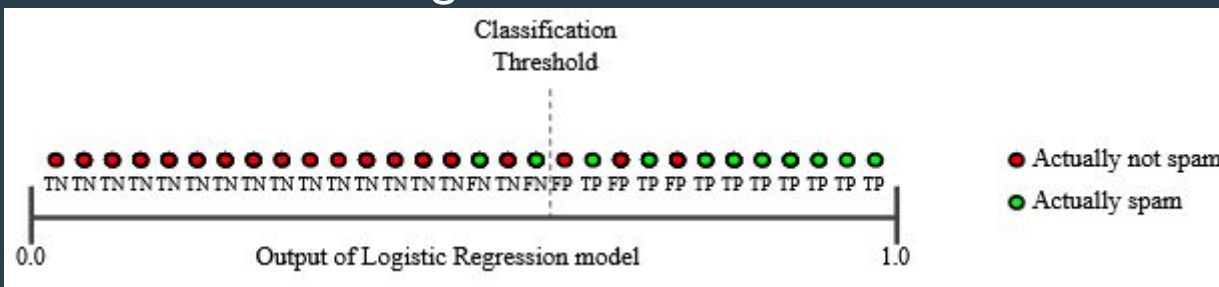
True Positives (TP): 7	False Positives (FP): 1
False Negatives (FN): 4	True Negatives (TN): 18

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{7}{7 + 1} = 0.88$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{7}{7 + 4} = 0.64$$

number of false positives decreases, but false negatives increase - precision increases, while recall decreases

- effect of decreasing the classification threshold



True Positives (TP): 9	False Positives (FP): 3
False Negatives (FN): 2	True Negatives (TN): 16

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{9}{9 + 3} = 0.75$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{9}{9 + 2} = 0.82$$

false positives
increase, and false
negatives decrease -
precision decreases
and recall increases

Sensitivity vs Specificity

- Sensitivity (True Positive Rate)
 - measures the proportion of positives that are correctly identified
 - how well a test can identify true positive
 - i.e., the proportion of those who have some condition (affected) who are correctly identified as having the condition
 - $TPR = \frac{TP}{TP+FN}$

Sensitivity vs Specificity

- Specificity (True Negative Rate)

- measures the proportion of negatives that are correctly identified
- how well a test can identify true negatives
- i.e., the proportion of those who do not have the condition (unaffected) who are correctly identified as not having the condition

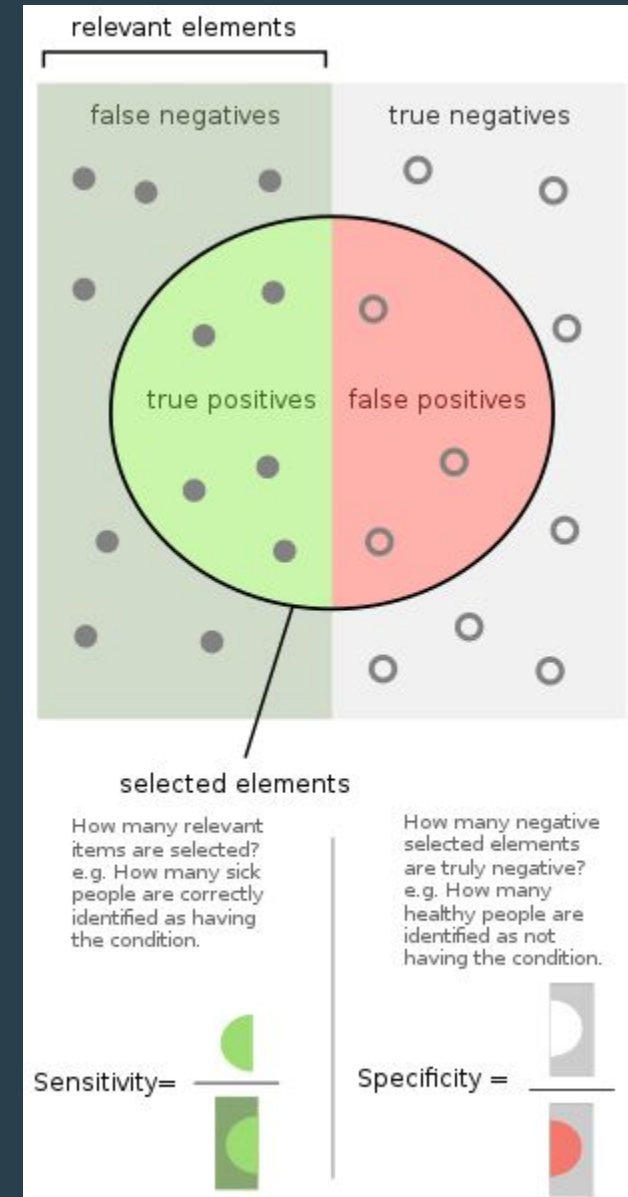
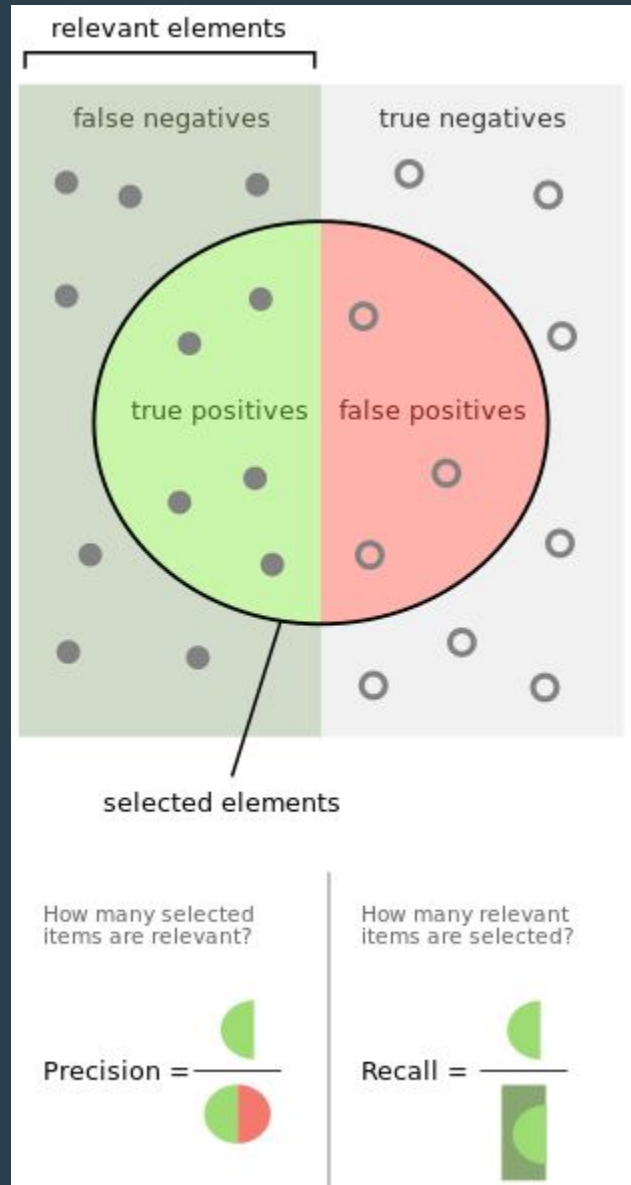
- $TNR = \frac{TN}{TN+FP}$

- F1 score

- in binary classification, F1 score is the harmonic mean of the precision and recall
- measure of a test's accuracy

- $F_1 = \frac{2}{recall^{-1} + precision^{-1}} = 2 \cdot \frac{precision \cdot recall}{precision + recall} = \frac{TP}{TP + \frac{1}{2}(FP + FN)}$

Precision vs Recall



Error Measurements

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Recall \text{ or } Sensitivity = \frac{TP}{TP + FN}$$

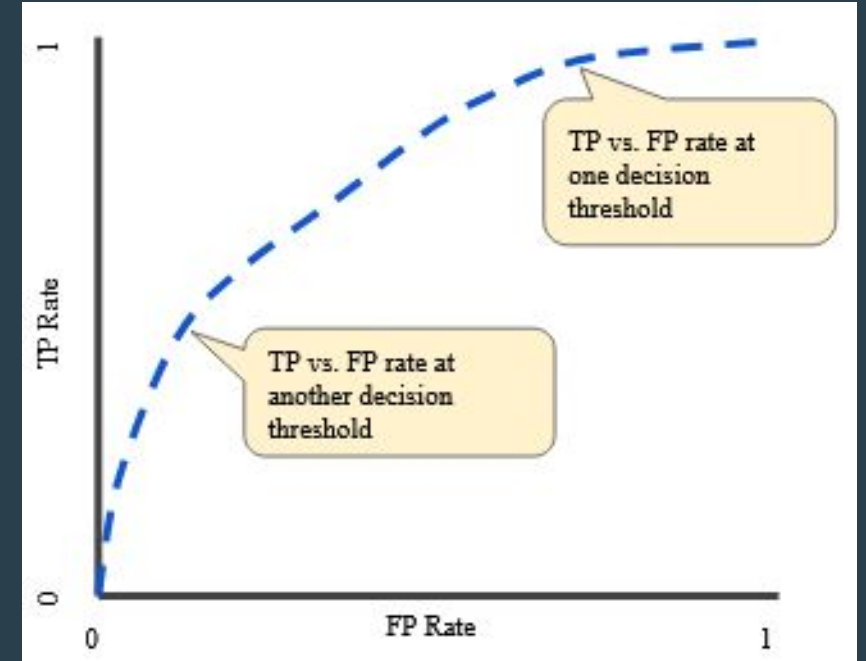
$$Precision = \frac{TP}{TP + FP}$$

$$Specificity = \frac{TN}{TN + FP}$$

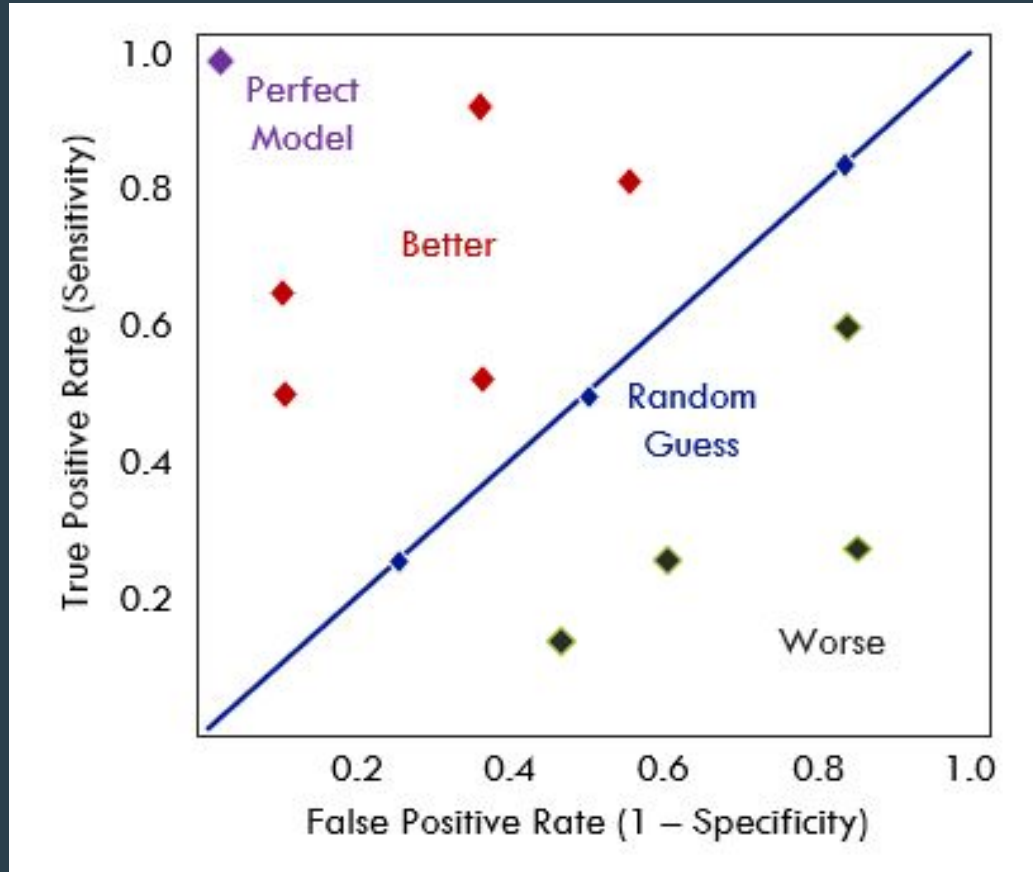
$$F_1 = 2 \cdot \frac{precision \cdot recall}{precision + recall} = \frac{2TP}{2TP + FP + FN}$$

ROC Curve

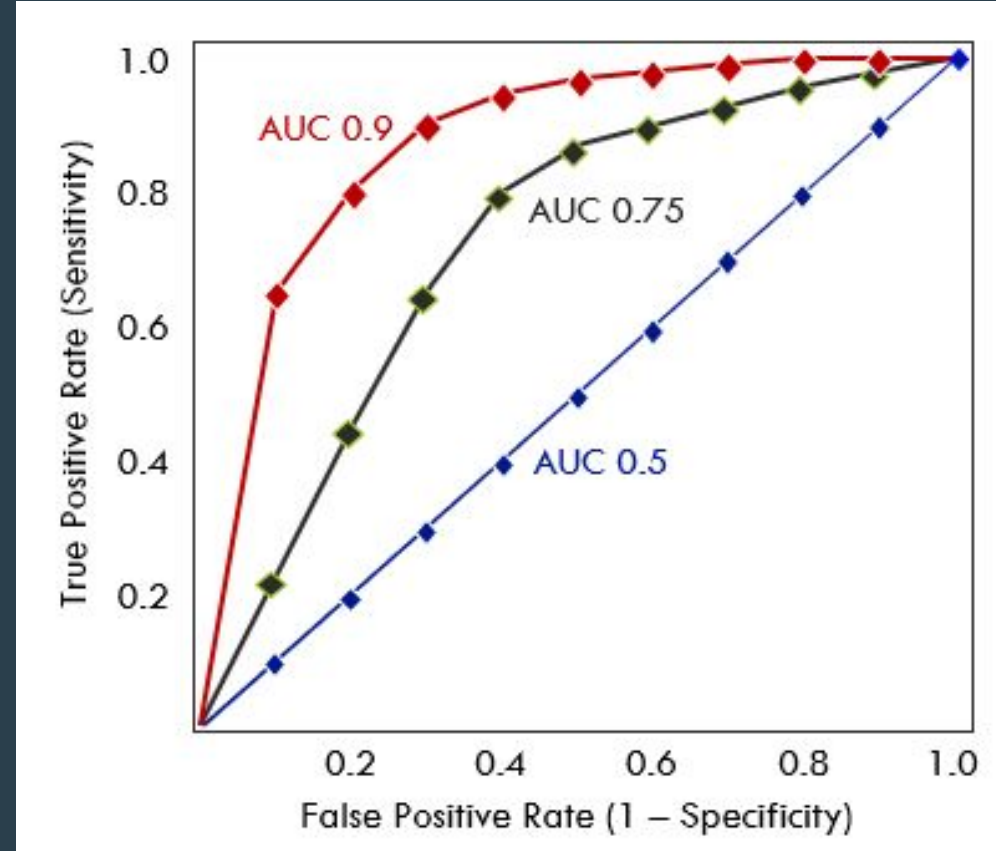
- receiver operating characteristic curve
- a graph showing the performance of a classification model at all classification thresholds
- the curve plots has two parameters:
 - True Positive Rate, $TPR = \frac{TP}{TP+FN}$
 - False Positive Rate, $FPR = \frac{FP}{TN+FP}$
- plots TPR vs. FPR at different classification thresholds
- lowering the classification threshold classifies more items as positive, thus increasing both False Positives and True Positives



ROC Curve



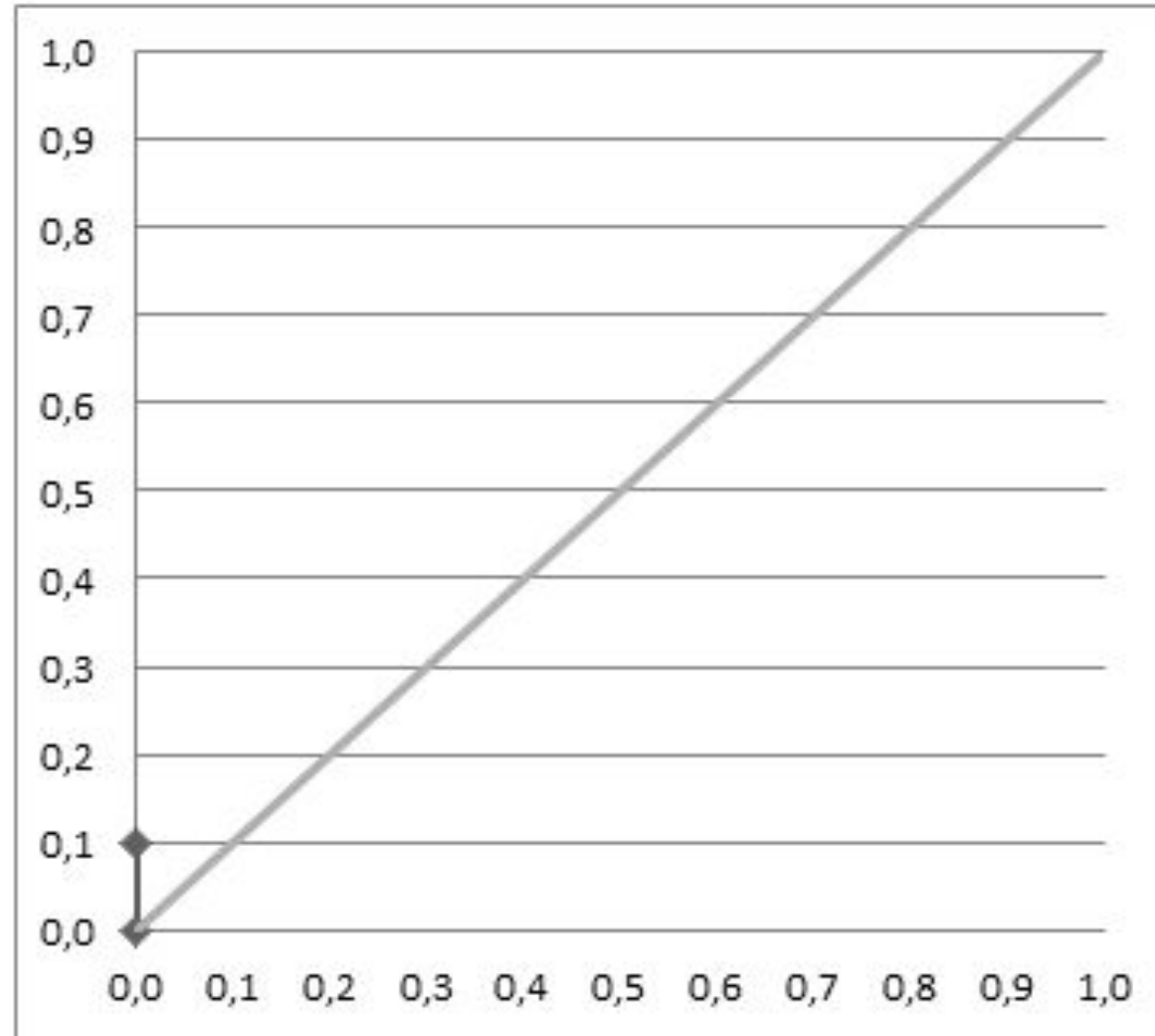
Evaluation of model at all possible thresholds



Measures total area under ROC curve

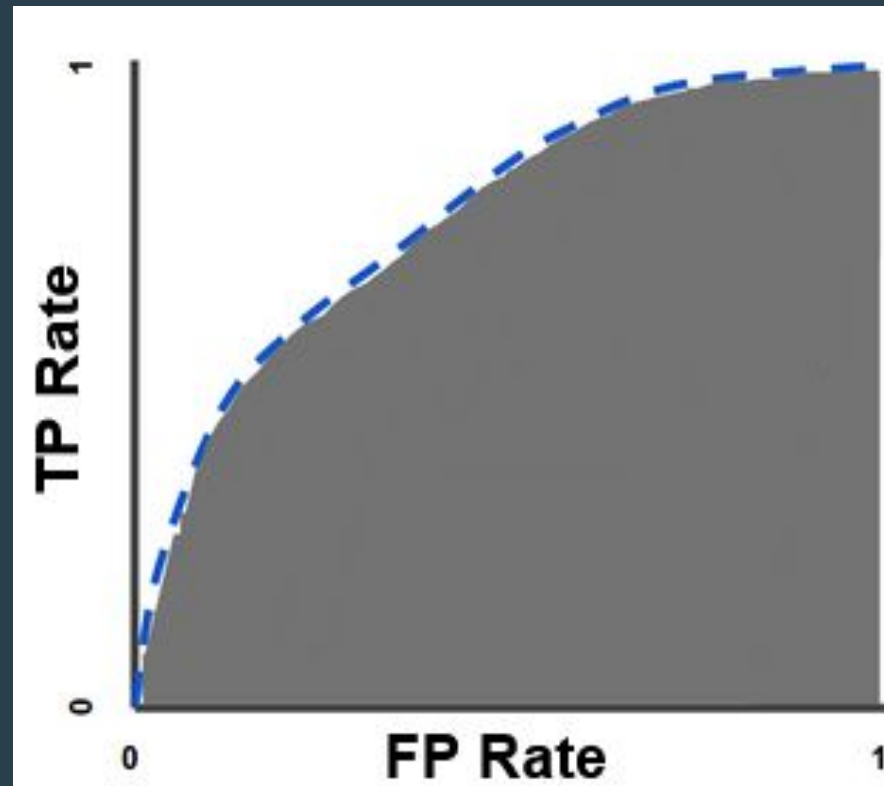
ROC Curve Creation

#	C	Score
1	P	0,9
2	P	0,8
3	N	0,7
4	P	0,6
5	P	0,55
6	P	0,54
7	N	0,53
8	N	0,52
9	P	0,51
10	N	0,505
11	P	0,4
12	N	0,39
13	P	0,38
14	N	0,37
15	N	0,36
16	N	0,35
17	P	0,34
18	N	0,33
19	P	0,3
20	N	0,1

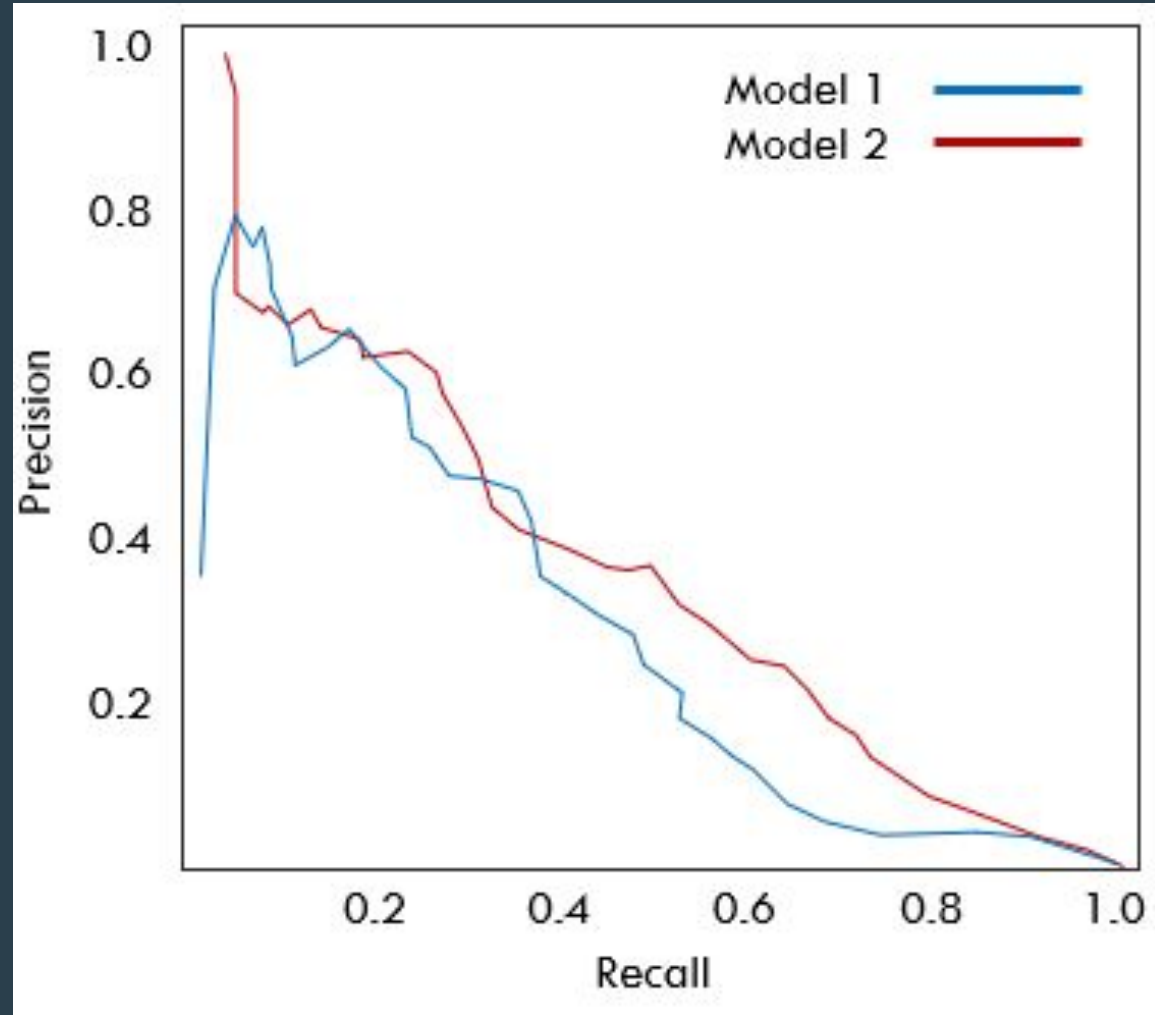


AUC Curve

- Area under the ROC Curve
- measures the entire two-dimensional area underneath the entire ROC curve (think integral calculus) from (0,0) to (1,1)
- provides an aggregate measure of performance across all possible classification thresholds. One way of interpreting AUC is as the probability that the model ranks a random positive example more highly than a random negative example
- 2 reasons of using AUC:
 - scale-invariant - measures how well predictions are ranked, rather than their absolute values
 - classification-threshold-invariant - measures the quality of the model's predictions irrespective of what classification threshold is chosen



Precision Recall Curve (PR Curve)



Measures trade-off between precision and recall

Multiple Class Error Metrics

	Predicted Class 1	Predicted Class 2	Predicted Class 3
Actual Class 1	TP1		
Actual Class 2		TP2	
Actual Class 3			TP3

$$Accuracy = \frac{TP1 + TP2 + TP3}{Total}$$

Most multi-class error metrics are similar to binary versions - just expand elements as a sum

Reference:

- <https://machinelearningmastery.com/calibrated-classification-model-in-scikit-learn/>
- <https://developers.google.com/machine-learning/crash-course/classification/accuracy>
- <https://developers.google.com/machine-learning/crash-course/classification/precision-and-recall>
- <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc>
- <https://machinelearningmastery.com/roc-curves-and-precision-recall-curves-for-classification-in-python/>